Roshan Balaji

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http://roshanmsb.github.io

Professional Experience

Nov 2025 – · · · · **Open Source Contributor**
OpenFold Initiative

May 2025 – Nov 2025 **Research Intern,** with Prof. Sarath Chandar, Mila-Quebec AI Institute
Goal Aware Retrosynthesis Planning

June 2022 – July 2022 **Teaching Assistant,** ACM – RBCDSAI Summer School
Lab sessions on mathematical foundations of data science

June 2020 – Nov 2020 **Research Intern,** Boltzmann Labs
Protein-ligand binding site prediction, De novo molecule generation

Education

2021 – · · · · **Ph.D., Indian Institute of Technology Madras**
Prime Minister's Research Fellow
Wadhwani School of Data Science & Department of Biotechnology
Teaching Assistant: Big Data and Biological Networks; Multivariate Data Analysis; Industrial AI Laboratory; Programming for Biological Engineers; Bioprocess Control; Current Topics in Synthetic Biology

2017 – 2021 **Bachelor of Technology, National Institute of Technology Calicut**
Department of Biotechnology
CGPA: 9.08 Gold Medalist
Thesis title: *Cancer subtype prediction using histopathology images*

Research Publications

Journal Articles and Preprints

1. **R. Balaji**, J. Bobby, and N. P. Bhatt, "Functional groups are all you need for chemically interpretable molecular property prediction," *Journal of Chemical Information and Modeling*, 2026. DOI: 10.1021/acs.jcim.5c02769.

Conference Proceedings and Presentations

1. A. Sinha, **R. Balaji**, N. P. Bhatt, et al., "Bica: Effective biomedical dense retrieval with citation-aware hard negatives," in *Proceedings of the AAAI Conference on Artificial Intelligence*, Oral (Top 5%), vol. 40, 2026, pp. 33 010–33 018.
2. K. Chitsaz, **R. Balaji**, Q. Fournier, N. P. Bhatt, and S. Chandar, "Novomolgen: Rethinking molecular language model pretraining," in *ICML 2025 Generative AI and Biology (GenBio) Workshop*, 2025. URL: https://openreview.net/forum?id=5OPItDY07C.

Miscellaneous Experience

Professional Service

- **Organizer and Program Chair**, GCLR Workshop 2023 & 2026 at AAAI Conference
- **Reviewer**, International Conference on Intelligent Systems for Molecular Biology (ISMB)